8.1 A 48Gb/s/lane 1.24Tb/s/mm UCIe-Compliant Die-to-Die Link Over 30mm Standard Package

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Abstract

This work demonstrates a UCIe-S compliant die-to-die PHY at 48Gb/s/lane across 16 lanes, achieving 1.24Tb/s/mm shoreline BW density over a 30mm organic package at 1.2pJ/b, extendable to 56GT/s (1.13pJ/b). Circuit innovations include an impedance-invariant TX CTLE, high-swing N/N driver, compact resonant clocking, double-tail latch w/ improved setup/hold times, and common-mode supply-noise rejection. Compared to prior UCIe-S, it achieves 3× higher data rate and 2.8× higher BW density.

As system complexity grows, monolithic SoC scaling faces challenges in integration, die size, and technology heterogeneity, motivating chiplet disaggregation that improves yield and cost while enabling IP reuse and heterogeneous integration. Die-to-die (D2D) interconnects with low latency, high bandwidth (BW) density and energy efficiency (EE) are therefore critical. The Universal Chiplet Interconnect Express (UCIe) standard [1] enables interoperability between dies from various vendors over multiple protocols (4-64Gb/s). This work demonstrates a UCIe-compliant physical link (PHY) over standard organic package (pkg), or UCIe-S, operating at the 48Gb/s/lane across all 16 lanes, and further extending to 56Gb/s. This was enabled by several circuit-level innovations: an impedance-invariant passive TX continuous-time linear equalizer (CTLE), a high-swing NMOS-over-NMOS (N/N) driver, coupling-canceled compact resonant clock (ck) distribution, a high-sensitivity double-tail latch (DTL) with improved setup/hold times (t_{SU}/t_H), and common mode (CM) supply-noise (SN) rejection in the RX using the track (trk) signal.

Figure 8.1.1 shows a block diagram of the UCIe-S PHY, consisting of 16 data lanes, quadrature (quad) forwarded clocks (ck_{fw}) at ¼ rate, and a trk/valid signal between the two dies. The dies are mounted on a 5-2-5 organic 2D standard package substrate with 30mm channels routed in two layers, as per UCIe specification. A differential (diff) ¼-rate external reference ck is supplied to each die. Quad phases generated by a coupled-resonator quad hybrid (QH) [2] drive two 8b phase interpolators (PIs) which provide the rotated ck to the 16 TX data lanes and the two quad ck_{fw}, respectively. The RX employs a matched architecture, with both TX and RX incorporating independent per-channel de-skew to compensate for lane-to-lane skew.

A passive CTLE, at the TX output or RX input, can equalize channel loss w/o power overhead when the TX driver directly drives the CTLE impedance (Z). Conventional passive CTLEs [3], however, exhibit strong frequency-dependent Z variation (~150% for 5dB peaking) at the pad, leading to reflections and ISI. A pole in the impedance response aligned with the CTLE zero ($-1/R_1C_1$ in Fig. 8.1.2) causes larger variation at higher equalization gain. This work mitigates the Z-variation by augmenting the series R_1–C_1 network with a shunt R_2–L_2 branch, which introduces an additional zero at $-R_2/L_2$ in both gain and impedance responses. The proposed CTLE therefore achieves higher peaking from two low-frequency zeros while maintaining near-constant output impedance through a compensating low-frequency pole–zero pair (Fig. 8.1.2). While the Z-invariance and peaking benefit holds both in the TX and RX configuration, the CTLE is implemented at the TX side in this work. The additional CTLE inductor is collocated with the TX BW-extension inductor, as the overall response is largely insensitive to their coupling, eliminating area overhead. Compared to a feedforward equalizer (FFE), the CTLE avoids extra data-path power and reduces TX driver power by a factor of $1+\alpha(1-\alpha)\times(R_R/R_T)$, where R_R, R_T and α are RX termination, TX term. and FFE coefficient, respectively. The CTLE also improves eye width (EW) through an enhanced high-frequency response (Fig. 8.1.2). Moreover, a TX FFE does not shape unmodulated low-frequency SN. However, the CTLE attenuates it by its equalization gain, making it beneficial for single-ended (SE) voltage-mode (VM) drivers. While the CTLE, tunable through R_1/R_2, is sufficient, one configurable pre/post-cursor FFE tap (not both as [1]) is implemented but unused.

To reduce ck distribution power and achieve better lane-to-lane matching, the active circuits of all 16 lanes are clustered into two groups (Fig. 8.1.7), with the ck driven symmetrically from a central location. TX outputs are routed to the pads through matched 200µm on-die T-lines, which act as extensions of package channels with negligible loss. Resonant clocking in the TX ck network halves power, with an option to switch to non-resonant mode below 12GHz using center-tap switches to support lower UCIe data rates (Fig. 8.1.3). Given the stringent bump-map area, multiple I/Q inductor pairs are impractical; instead, the global ck with a single I/Q inductor drives all 16 lanes w/o repeaters, enabled by lane clustering. The global ck for ck_{fw} and track lanes, with lighter loading, is distributed in non-resonant mode. A single I/Q pair with matched inductor is challenging to accommodate; so, the I/Q inductors are overlapped, with an '8-shaped' [4] turn introduced in one winding to cancel mutual coupling that would otherwise cause large I/Q gain and phase errors (Fig. 8.1.3). Each TX lane employs per-lane de-skew to compensate up to 0.5UI lane-to-lane variation as required by UCIe-S spec. To meet this range efficiently, a "sub-UI PI" is used instead of inverter-based delay lines. The sub-UI PI is implemented with interpolating strength set to half of the main path, incurring ~40% higher power only in the first-stage local ck buffer. Furthermore, since the inner lanes away from the die edge incur an additional ~6ps delay relative to the outer lanes, the sub-UI PI interpolates a leading phase for inner lanes and a lagging phase for outer lanes (Fig. 8.1.3). Using separate lead/lag sub-UI PIs reduces its power by ~40% compared to supporting both options in every lane.

The TX 4:1 serializer (Fig. 8.1.3) uses a combinational 1UI pulse generator (PG) and directly combines 4UI data at the high-BW TX output node. Unlike [2], three inverter stages are inserted between the PG and driver to reduce PG power and ck loading. As the pre-drivers buffer a 1UI pulse padded with zeros over the remaining three UIs, they tolerate moderate fanout while remaining ISI-free, unlike full-rate data paths where combining is done before the output. A VM driver is adopted for efficiency, but conventional VM drivers face a tradeoff between low swing in N/N drivers [2] and high power in linearized PMOS-over-NMOS (P/N) drivers, which require upsizing of the driver and preceding stages [5]. In this work, a high-swing N/N driver is introduced that operates at nominal supply and employs a PMOS resistor (~⅓ of the termination resistance) at the pull-up NMOS drain. When the pull-up branch is activated, the voltage across this PMOS resistor pushes the NMOS into the linear region, improving linearity. Simulations show 2.5× higher EE than linearized P/N drivers and 1.5× greater swing than low-swing N/N drivers. The final pre-driver stage uses a higher supply V_{DDH} than IO supply V_{DD}, within the process-nominal supply range, to further aid linearization and mitigate pulse-width distortion inherent to N/N drivers [2].

The ck distribution in the matched RX architecture (Fig. 8.1.4) receives two quad phases of the ck_{fw} from the TX, converts them to diff and further amplifies the four quad phases. Similar to the TX, a resonant buffer with coupling-cancelling I/Q inductors distributes the ck to all 16 RX lanes, followed by per-lane buffering and de-skew. At the RX lane, the SE input data from the TX is converted to diff and amplified through high-BW inverter-based Cherry–Hooper (CH) stages, where g_m stages drive shunt-feedback transimpedance amplifiers; this RX amplifier chain delays the data-path to match the ck-path timing while amplifying the data with minimal jitter amplification. As the TX signal has a low CM voltage (~0.1V), the input g_m stage uses a PMOS device M_p biased by an NMOS current source M_n. A closed-loop DC offset correction (DCOC) minimizes the diff output CM mismatch of the RX data amplifier chain by injecting current into a resistor R at the input. The current injected into R raises the gate voltage of M_p and keeps it in saturation, while a parallel capacitor C extends the BW.

To improve link performance in SE UCIe-S, the RX employs CM noise rejection using the trk signal (Fig. 8.1.4). The low-frequency TX SN, common to both the trk and data paths, is extracted from the trk path using an RC filter. This filtered signal ($V_{B\text{-}DATA}$) is distributed to all 16 lanes and applied to the M_n devices that bias the respective input PMOS M_p of the CH amplifier, thereby canceling the low-frequency TX-SN in the data path. The amplified and delay-matched diff output then drives four slices of a ¼-rate cascaded DTL. The 2-ck-phase DTL [6] improves ck-to-Q delay and sensitivity, which are further enhanced by cascading with a ¼-sized 1-ck-phase DTL driven by a complementary ck phase. This cascaded DTL achieves <20m$V_{pk\text{-}pk,diff}$ simulated sensitivity at 56Gb/s or, equivalently, <2ps simulated t_{SU}/t_H with a 400m$V_{pk\text{-}pk}$-swing diff signal, improving link timing margin.

Fabricated in 22nm FinFET CMOS with 112.65µm bump pitch, the UCIe-S PHY core occupies 1.24×0.72mm² (Fig. 8.1.7). Two dies were routed across a 30mm on-pkg channel, mounted on a custom PCB supplying external ck and DC power supplies. An on-die PRBS generator produced PRBS15 data at the TX, and the 1/8-rate deserialized RX output was routed off-chip to an external BERT.

For standalone TX characterization, a die on the same package (Fig. 8.1.7) drove a 10mm channel to a high-speed probe, with the probe, cable and channel producing ~5.5dB Nyquist loss, comparable to the 30mm UCIe-S link, for representative TX eye monitoring. With the passive TX CTLE at its optimal setting, the measured 56Gb/s eye was 112mV/10ps at BER=1e-12, while the least-peaking setting (~3dB lower peaking) degraded the eye to 80mV/7.5ps, showing CTLE effectiveness (Fig. 8.1.6).

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The link operates at 48Gb/s per lane with all 16 lanes active. At V_{DD}=0.7V (V_{DDH}=0.85V for TX pre-driver), the worst-case bathtub curve across 16 channels shows a 1e-12 EW of 0.26UI (Fig. 8.1.5). BER was measured directly to 1e-12; extrapolation shows 0.15UI at 1e-15. Bathtub curves were obtained by measuring the PI output delay across code and mapping it to UI delay. The link achieves UCIe-S-compliant 48Gb/s/lane data rate at an energy efficiency of 1.2pJ/b (0.45, 0.69, and 0.06pJ/b in TX, RX, and PI+QH, respectively; a PLL is not integrated in this work). The link also extended operation to 56Gb/s with all 16 lanes active, at an EE of 1.13pJ/b (0.44-TX, 0.63-RX, 0.06-PI+QH) with the same V_{DD}/V_{DDH}. Representative bathtub curves for four lanes are shown in Fig. 8.1.5, with a worst-case 1e-12 EW of 0.23UI. Compared to prior D2D NRZ links (Fig. 8.1.6) over organic pkg [7-11], this work achieves the highest demonstrated data rate of 56Gb/s/lane, similar to [9], but with better EE despite using a less advanced process node. The link delivers a high shoreline density of 1.45Tb/s/mm and is UCIe compliant with 2-layer package routing at 3x the channel length compared to the 4nm design in [10], which uses 3 package layers. Compared to the 16Gb/s UCIe-S compliant design in 3nm CMOS [11], this 48Gb/s UCIe-S compliant design in 22nm FinFET achieves 3× higher data rate and 2.8× higher shoreline density.

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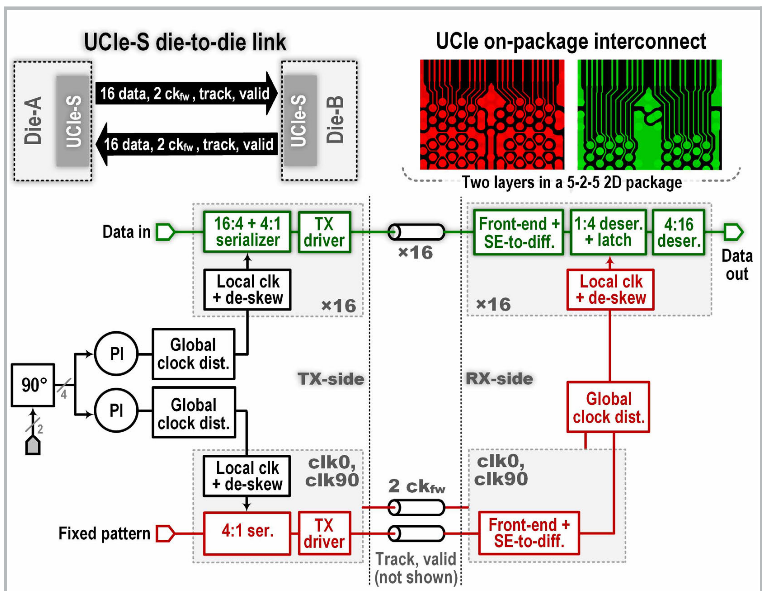


Figure 8.1.1: UCIe-S die-to-die system over ultra-dense package interconnect (top) and transceiver architecture highlighting data/clock paths (bottom).

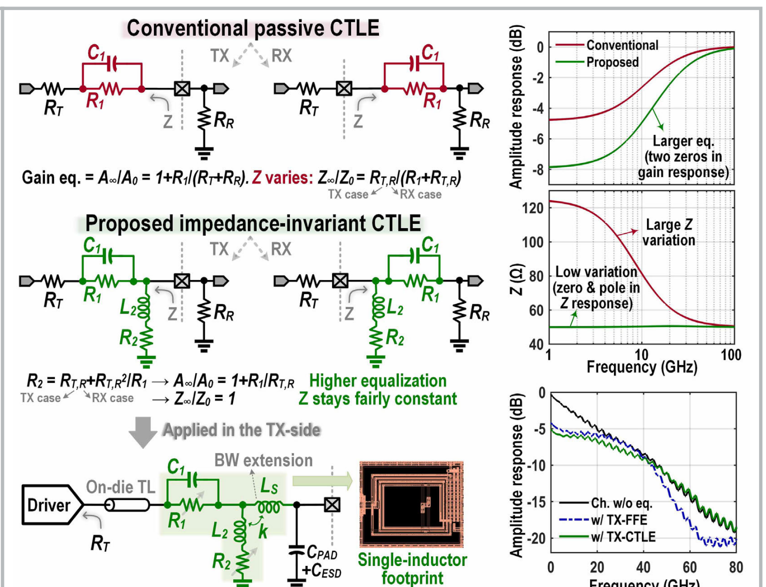


Figure 8.1.2: Proposed impedance-invariant passive CTLE and representative simulations illustrating impedance invariance and channel equalization.

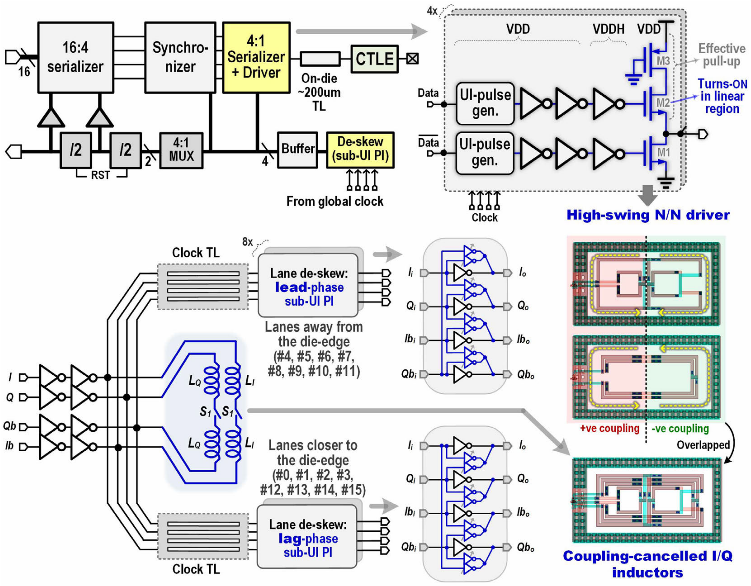


Figure 8.1.3: TX lane with proposed high-swing N/N driver (top); clock distribution with per-lane sub-UI PI and global I/Q overlapped inductors (bottom).

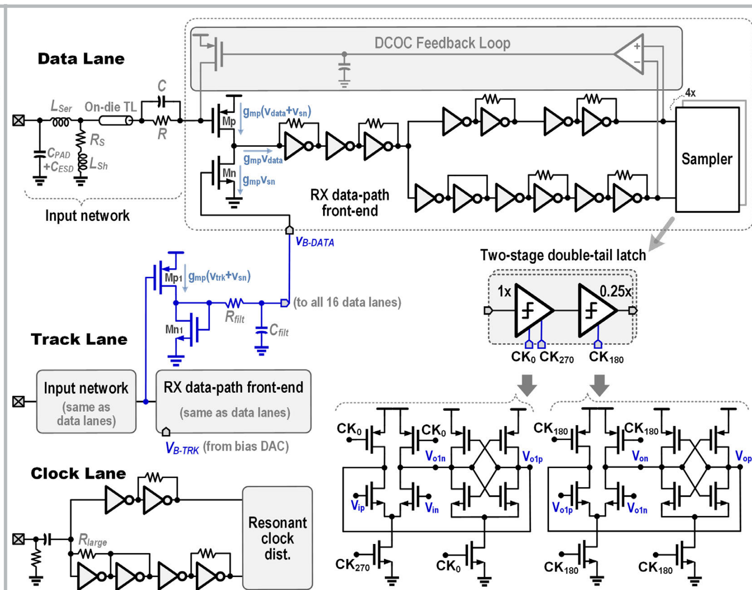


Figure 8.1.4: RX data, track, and clock path architecture with two-stage latch and common-mode supply-noise rejection using the track lane.

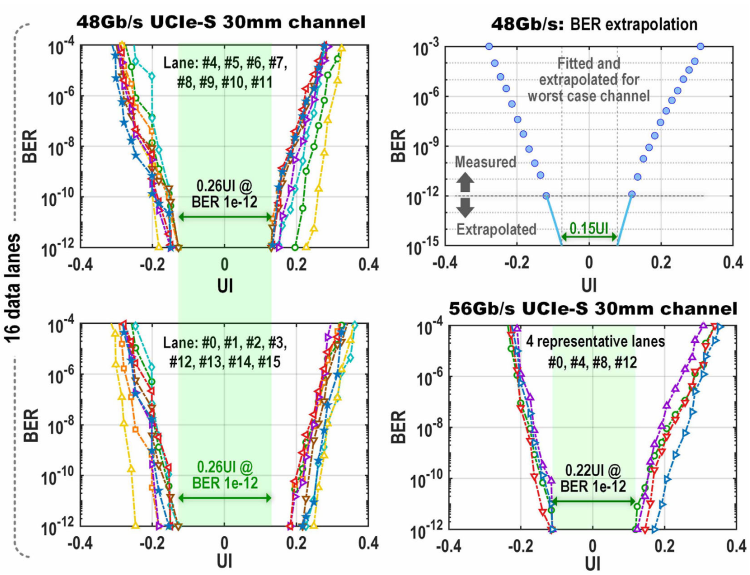
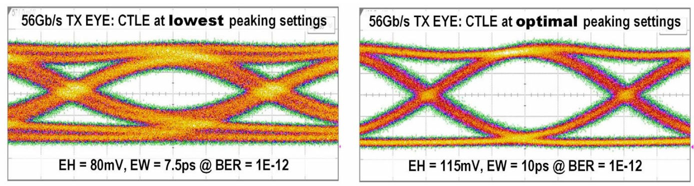


Figure 8.1.5: Measured bathtub plots for UCIe-S D2D interconnect over 30mm standard organic package: 48Gb/s (left, top-right) and 56Gb/s (bottom-right).



	Poulton, JSSC'19 [7]	McCollough, ISSCC'21 [8]	Poon, JSSC'22 [9]	Gangasani, ISSCC'22 [12]	Seong, ISSCC'24 [10]	Vandersand, VLSI'25 [11]	This Work	
Process	16nm	7nm	7nm	5nm	4nm	3nm	22nm	
Standard	Custom	Custom	Custom	Custom	Custom	UCIe-S	UCIe-S	Custom*
Package type	Organic	Organic	Organic	Organic	Organic	Organic	Organic	
Modulation	NRZ	NRZ	NRZ/PAM4	PAM4	NRZ	NRZ	NRZ	
Bump pitch [μm]	-	130	-	-	130	130	112.65	
Data rate [Gb/s]	25	40	56 (NRZ)/ 112 (PAM4)	113	48	16	48Gb/s	56Gb/s
Channel length [mm]	10 (80 off-pkg)	20	5-30	5-80	10	5-25	30mm	
Shore-line BW [Tb/s/mm]	0.200	0.450	0.870	0.460	1.850	0.448	1.243	1.450
Energy efficiency [pJ/b]	1.17	1.7	1.24	1.55	0.67†	0.52	1.2pJ/b†	1.13pJ/b†

*UCIe-S compliant at 48 Gb/s; 56 Gb/s is an extended data-rate (not part of the current UCIe-S standard). †A PLL is not integrated in this work.

Figure 8.1.6: Measured TX eye at 56Gb/s with CTLE at lowest and optimal peaking (top); performance comparison to custom and UCIe-S standard on-package links (bottom).

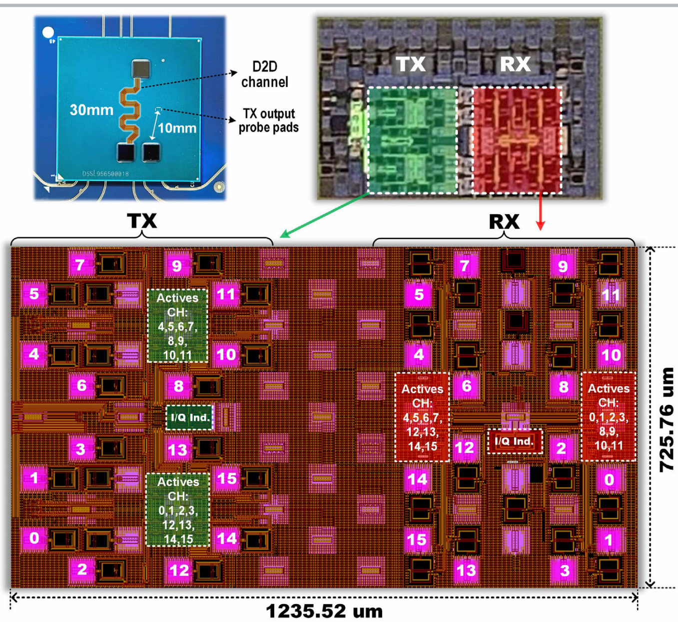


Figure 8.1.7: Package photograph and die micrograph (top); TX/RX floorplan (bottom).

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